

PARTH MAHENDRA PURI

AI ENGINEER

B.E. Artificial Intelligence & Data Science (AI & DS)

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Detail-oriented and organized AI engineer with a strong foundation in Machine Learning, Deep Learning, Generative AI and Agentic AI workflow. Skilled in developing innovative solutions using Multi Modal LLMs and State of the art technologies.

EXPERIENCE

Artificial Intelligence (AI) Engineer at Verdantis (Ultria, Verdantis), Mumbai.

Mar 2025 - Present

- Developed AI systems by leveraging top-tier generative models Google Gemini 2.5 Pro / Flash, Open AI GPT-4.1 / GPT-5, Anthropic Claude 3.5 Sonnet, Anthropic Claude 3.7 Sonnet, Anthropic Claude 4, 4.5 Sonnet, and Anthropic Claude 4, 4.1 Opus to deliver scalable, enterprise solutions.
- Designed Agentic Workflows and end-to-end Generative AI pipelines for autonomous, context-aware task execution and developed robust AI Prediction using Generative AI, increasing SMEs (Subject Matter Experts) work efficiency in the MRO industry by 95%.
- Implemented AI Governance by designing Workflows with strict Human-in-the-Loop (HITL) validation gates, ensuring Experts verified all automated outputs before deployment, Engineered Responsible AI by reducing hallucinations by 40% and preventing the generation of non-factual corporate data.

Machine Learning Engineer at Health India Insurance TPA Pvt Ltd, Mumbai.

Nov 2024 - Mar 2025

- Developed and trained Machine Learning and Deep Learning models to automate processes and enhance operational efficiency 70%.
- Built data extraction pipelines leveraging Generative AI and worked extensively with Large Language Models (LLMs), focusing on fine-tuning LLMs and hyperparameter optimization, achieving over 91% accuracy in UAT testing.
- Worked with various pre-trained models like BERT, Llama (3.2-3B & 1B), Mistral (3B & 7B), Deepseek (7B, 8B, 14B), YOLO, LSTMs, GRUs, as well as applications in Image Processing, Azure OCR, document reconstruction, text classification, and transformers. Also Built RAG pipelines with MongoDB and Langchain. Ensured PII security and compliance by deploying an on-premise pipeline with zero external data transfer.

AI Engineer Intern at Sharv Auto Tech Solutions Private Limited, Mumbai.

May 2024 - Nov 2024

- Enhancing Natural Language Processing capabilities using Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), including finetuning LLMs. Working with SQL Database and Fast API for backend development.

AI Engineer Intern at Fafadia Tech, Mumbai.

May 2023 - Apr 2024

- Worked on Pretrained Models, Similarity Modeling, and Transfer Learning. Developed Machine Learning and Deep Learning Models using Python.

PROJECTS

Research Assistant Agent

Jan 2026 - Mar 2026

- Architected a fully autonomous, end-to-end research pipeline using LangGraph with a Supervisor-Worker multi-agent architecture, implementing a self-evaluating feedback loop that dynamically assesses research quality in real-time and autonomously triggers refined search iterations with zero manual intervention. Leveraged Agentic Function Calling for intelligent task decomposition and dynamic workflow routing, enabling goal-directed autonomous execution across specialized worker agents.
- Engineered 100% a production-grade FastAPI async REST API with PostgreSQL for stateful agent orchestration at scale.

Retroflex: Uncovering Visual Equivalences through Reverse Image Recon.

May 2023 - Apr 2024

- Project focuses on Reverse Image Search (RIS) using Content-Based Image Retrieval (CBIR) techniques.
- It leverages Deep Learning methods, including Auto-encoders and Convolutional Neural Networks (CNNs), for advanced feature extraction from large datasets, reducing search latency by 40%, significantly enhancing user experience and system performance.
- Published at IEEE 2024 Delhi Section (INDIACom-2024; International Conference on Computing for Sustainable Global Development).

SKILLS

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|----------------------------|-------------------|-----------------|--------------|-----------------|
| • Python. | • Generative AI. | • Docker | • LangChain. | • GCP |
| • Artificial Intelligence. | • Agentic AI | • Vector Store. | • LangGraph | • Git & Github. |
| • Machine Learning. | • LLM RAG Model. | • Hugging Face. | • Fast API. | • Linux. |
| • Deep Learning. | • LLM Finetuning. | • Tensorflow. | • AWS. | |

AWARDS RECEIVED

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| • Best Team Award (Team AI) at Verdantis (Ultria, Zycus) 2025. | • Winner of the VCET'S National Level Project Showcase Competition 2023. |
| • Winner of Code-O-Fiesta Hackathon 2022. | • Runners-up at VNPS 2024. |
| • Winner of Code-O-Fiesta Hackathon 2023. | |

EDUCATION

B.E. in Artificial Intelligence and Data Science.

Jun 2020 - May 2024

University of Mumbai, Vidyavardhini's College Engineering and Technology, Mumbai - 8.04 CGPA

HSC - Bhartiya Vidya Bhavans College (Andheri).

May 2018 - May 2020

CERTIFICATION AND PUBLICATION

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| • Published Research Paper named "Retroflex: Uncovering Visual Equivalences through Reverse Image Recon," 2024 11th INDIACom, New Delhi, India, 2024. IEEE.explore | Mar 2024 |
| • Software Engineer Intern, HackerRank. HackerRank | Feb 2024 |
| • Data Science Intern, Mumbai Metropolitan Region Development Authority (MMRDA). | Jun 2022 - Jul 2022 |
| • Generative AI on GCP: Deep Dive Into Generative AI Studio. InfosysSpringBoard | Aug, 2024 |
| • Machine Learning with TensorFlow (InfosysSpringBoard). | Jul, 2024 |